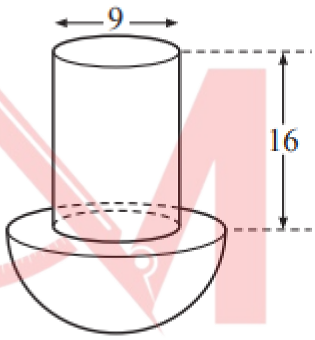
Mensuration

Worksheet 3

Composite Solids

1. W20/qp22/q4(a)(i)(ii)

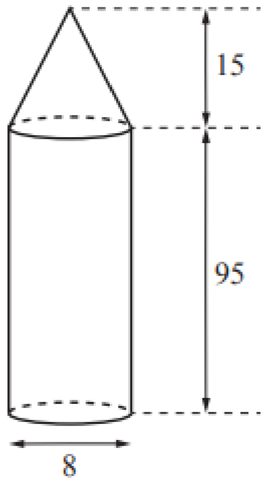


The diagram shows a solid formed by joining a cylinder to a hemisphere. The diameter of the cylinder is 9cm and its height is 16cm.

i. The volume of the hemisphere is equal to the volume of the cylinder. Show that the radius of the hemisphere is 7.86cm, correct to 2 decimal places. [4]

ii. Calculate the total surface area of the solid. [3]

2. W19/qp21/q4



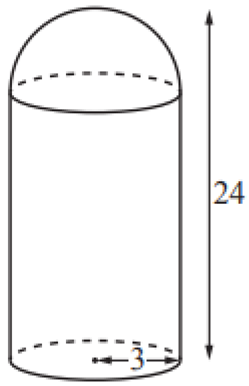
The diagram shows a gate post. It is made in the shape of a cylinder with a cone on top. The cylinder and the cone each have diameter 8cm. The height of the cylinder is 95cm and the height of the cone is 15cm.

a. Calculate the volume of the gate post. [3]

b. Show that the total curved surface area of the gate post is $2580cm^2$, correct to 3 significant figures. [5]

c. A geometrically similar gate post has a total height of 150cm.
Calculate the total curved surface area of this gate post. [2]

3. W18/qp21/q9(a)



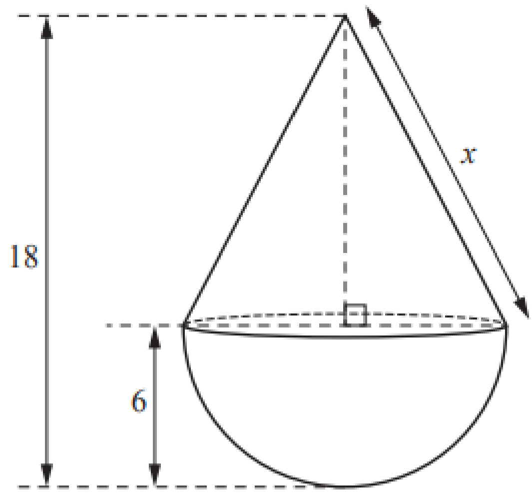
The diagram shows lamp A. It is made in the shape of a cylinder with a hemisphere on top. The radius of the hemisphere and the radius of the cylinder are both 3cm. The total height of the lamp is 24cm.

i. Show that the volume of lamp A is $650cm^3$, correct to 3 significant figures. [4]

ii. Calculate the curved surface area of lamp A. [3]

iii. Lamp B is mathematically similar to lamp A. The volume of lamp B is $450\ cm^3$. Calculate the total height of lamp B. [2]

4. W17/qp21/q8



The diagram shows solid A which is made from a hemisphere joined to a cone of equal radius. The hemisphere and the cone each have radius 6cm. The total height of the solid is 18cm.

a. Show that the slant height, x cm, of the cone is 13.4cm, correct to 1 decimal place. [2]

b. Calculate the total surface area of solid A. [3]

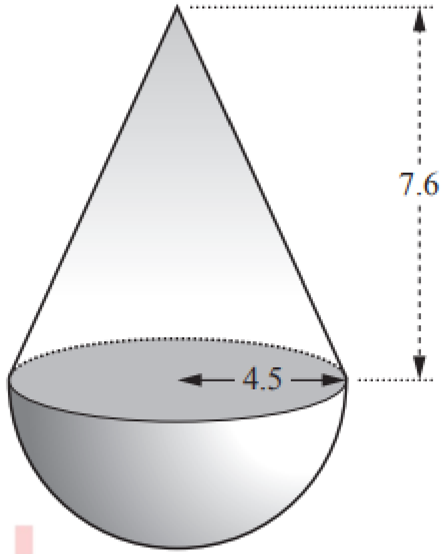
c. Calculate the volume of solid A. [3]

d. Solid A is one of a set of three geometrically similar solids, A, B and C. The ratio of the heights of solid A : solid B : solid C is 2 : 6 : 1.

   i. Calculate the surface area of solid B correct to 3 significant figures. [2]

   ii. Calculate the volume of solid C correct to 3 significant figures. [2]

5. S15/qp21/q4



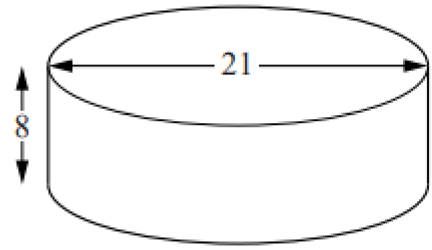
A solid is formed by joining a cone of radius 4.5cm and height 7.6cm to a hemisphere of radius 4.5 cm as shown.

a. Calculate the area of the circle where they are joined. [2]

b. Calculate the total volume of the solid. [2]

c. Another solid of the same type is made by joining a cone of radius 5cm and height h cm to a hemisphere of radius 5cm. The cone and hemisphere have equal volumes. Calculate the height of the cone.[2]

6. W12/qp22/q7



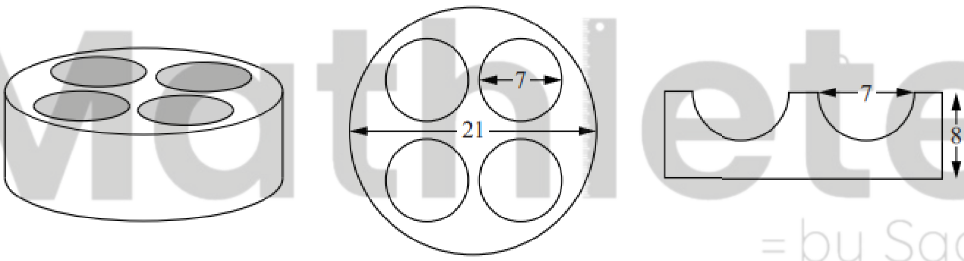
A cylindrical, open container has a diameter of 21 cm and height of 8 cm.

a.

i. Calculate the total external surface area of this container. [3]

ii. A manufacturer receives an order for 30 000 containers. He needs an extra 150 cm^2 of material for each container to cover wastage. Calculate the area of material needed to make these containers. Give your answer in square metres. [2]

b. A circular top that can hold 4 hemispherical bowls can be placed on the container.



Container and Top **Top** **Cross-section**

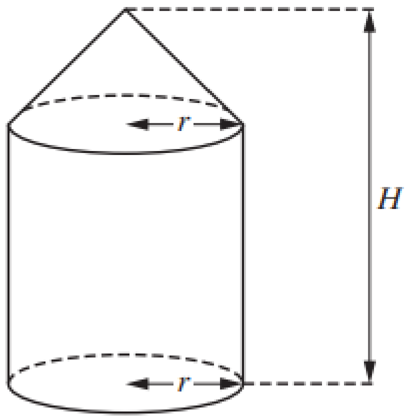
The top is a circle of diameter 21 cm with four circular holes of diameter 7 cm. A hemispherical bowl of diameter 7 cm fits into each hole. The cross-section shows two of these bowls.

i. Calculate the inside curved surface area of one of these hemispherical bowls. [1]

ii. Calculate the total surface area of the top of the container, including the inside curved surface area of each bowl. [3]

iii. With the top and the 4 bowls in place, calculate the volume of water required to fill the container. [3]

7. S11/qp22/q11



The solid above consists of a cone with base radius r centimetres on top of a cylinder of radius r centimetres. The height of the cylinder is twice the height of the cone. The total height of the solid is H centimetres.

a. Find an expression, in terms of π, r and H, for the volume of the solid. Give your answer in its simplest form. [3]

b. It is given that r = 10 and the height of the cone is 15 cm.

   i. Show that the slant height of the cone is 18.0 cm, correct to one decimal place. [2]

   ii. Find the circumference of the base of the cone. [2]

   iii. The curved surface area of the cone can be made into the shape of a sector of a circle with angle θº. Show that θ is 200, correct to the nearest integer. [2]

   iv. Hence, or otherwise, find the total surface area of the solid. [3]

8. W22/qp41/q1 (0580)

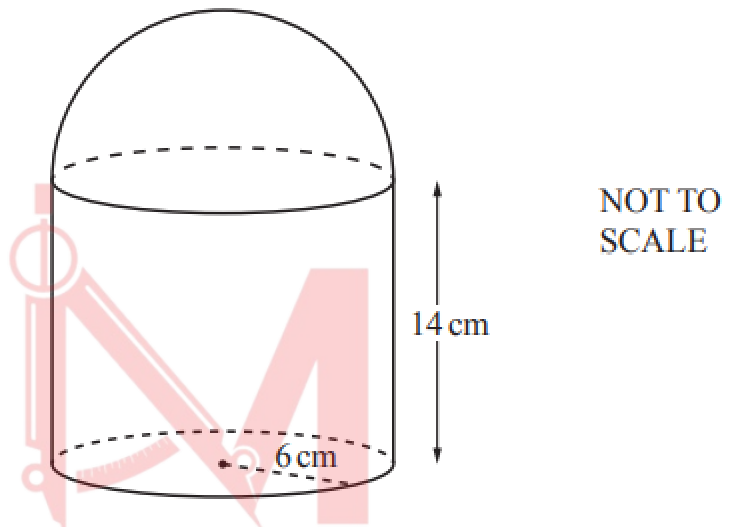
a. Calculate the volume of

i. a solid cylinder with radius 6cm and height 14cm, [2]

ii. a solid hemisphere with radius 6cm. [2]

b. The cylinder and hemisphere in part **(a)** are joined to form the solid in the diagram. The solid is made of steel and 1 cm^3 of steel has a mass of 7.85g.

i. Show that 1 cm^3 of steel has a mass of 0.00785kg. [1]



ii. Calculate the total mass of the solid. [2]

c. $2000\ cm^3$ of iron is melted down and some of it is used to make 50 spheres with radius 2 cm.

i. Calculate the percentage of iron that is left over. [3]

ii. The iron left over is then made into a cube

Calculate the length of an edge of the cube. [1]

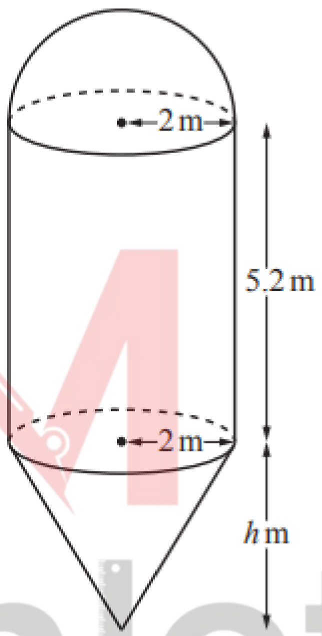
d. A solid cone has radius $3Rcm$ and slant height $9Rcm$. A solid cylinder has radius $x\ cm$ and height $7x\ cm$. The total surface area of the cone is equal to the total surface area of the cylinder. Given that $R = kx$,

find the value of k. [4]

9. W21/qp42/q7(a) (0580)

The diagram shows a container for storing grain. The container is made from a hemisphere, a cylinder and a cone, each with radius 2m. The height of the cylinder is 5.2m and the height of the cone is hm.

i. Calculate the volume of the hemisphere. Give your answer as a multiple of π. [2]



ii. The total volume of the container is $\frac{88\pi}{3}\ m^3$. Calculate the value of h. [4]

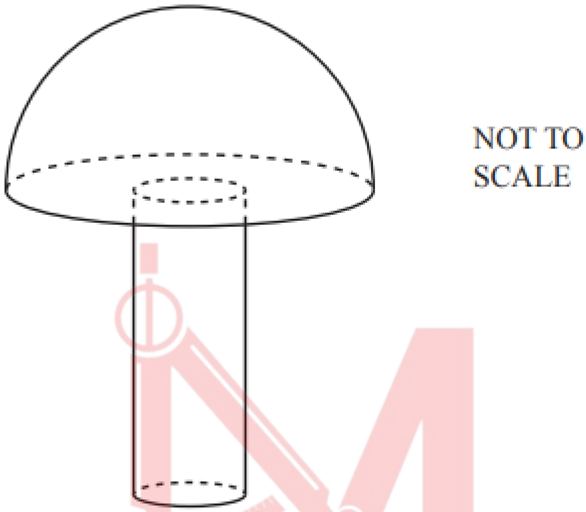
iii. The container is full of grain. Grain is removed from the container at a rate of 35000kg per hour. $1m^3$ of grain has a mass of 620kg. Calculate the time taken to empty the container. Give your answer in hours and minutes. [3]

10. S18/qp41/q6 (0580)

A solid hemisphere has volume $230cm^3$.

a. Calculate the radius of the hemisphere. [3]

b. A solid cylinder with radius 1.6cm is attached to the hemisphere to make a toy. The total volume of the toy is $300cm^3$.



i. Calculate the height of the cylinder. [3]

ii. A mathematically similar toy has volume $19200cm^3$. Calculate the radius of the cylinder for this toy. [3]

Answer Key

1. W20/qp22/q4(a)(i)(ii)

4(a)(i)	$\pi \times \left(\frac{9}{2}\right)^2 \times 16 = \frac{1}{2} \times \frac{4}{3} \times \pi \times r^3$	M2	M1 for $\pi \times \left(\frac{9}{2}\right)^2 \times 16$ oe or $\frac{1}{2} \times \frac{4}{3} \times \pi \times r^3$ oe
	$r^3 = \frac{3}{2} \times \left(\frac{9}{2}\right)^2 \times 16$ or $r = \sqrt[3]{\frac{3}{2} \times \left(\frac{9}{2}\right)^2 \times 16}$	M1	
	$r = 7.862...$	A1	
4(a)(ii)	1030 or 1040 or 1034.6 to 1035.1...	3	M1 for $\pi \times 9 \times 16$ oe M1 for $2 \times \pi \times 7.86^2$ oe or $3 \times \pi \times 7.86^2$ oe

2. W19/qp21/q4

4(a)	5030 or 5026 or 5027 or 5026.5 to 5027.2	3	M1 for $\frac{1}{3}\pi \times 4^2 \times 15$ M1 for $\pi \times 4^2 \times 95$ After 0 scored, SC1 for answer 20 100 or 20 110 or 20 106 to 20 109
4(b)	$[l =]\sqrt{15^2 + 4^2}\left(= \sqrt{241}\right)$	M2	M1 for $[l^2 =]\ 15^2 + 4^2$ oe
	Curved surface area $= 4\pi \times \sqrt{241} + \pi \times 8 \times 95$	M2	M1 for $4\pi \times their\sqrt{241}$ or $\pi \times 8 \times 95$
	=2582.3 to 2583.03=2580 **AG**	A1	
4(c)	4800 or 4797.5 to 4803.2	2	M1 for $\left(\frac{150}{95+15}\right)^2$ or $\left(\frac{95+15}{150}\right)^2$ soi

3. W18/qp21/q9(a)

9(a)(i)	$\pi \times 3^2 \times 21 + \frac{2}{3} \times \pi \times 3^3$	M3	**B1** for cylinder height = 21 **soi** **M1** for $\pi \times 3^2 \times$ *their* height **M1** for $\frac{2}{3} \times \pi \times 3^3$
	= 650.3[…] or 650.4	A1	
9(a)(ii)	452 or 452.3 to 452.4…	3	**M2** for $2 \times \pi \times 3^2 + \pi \times 6 \times 21$ or **M1** for $2 \times \pi \times 3^2$ or $\pi \times 6 \times 21$
9(a)(iii)	21.2 or 21.22 to 21.23…	2	**B1** for $\sqrt[3]{\frac{450}{650}}$ **soi** or $\sqrt[3]{\frac{650}{450}}$ **soi**

4. W17/qp21/q8

8(a)	$[x^2 =]\ 6^2 + 12^2$	M1	or $[x =]\sqrt{6^2 + 12^2}$
	$[x =]$ 13.41[6…] or 13.42	A1	
8(b)	478.7 to 479.4	3	**M1** for $\left[\frac{1}{2} \times\right] 4 \times \pi \times 6^2$ seen **M1** for $\pi \times 6 \times 13.4$ seen After 0 scored, **SC1** for consistent use of r = 3 in formula for [hemi]sphere and cone
8(c)	904.7 to 905 **nfww**	3	**M1** for $\left[\frac{1}{2} \times\right] \frac{4}{3} \times \pi \times 6^3$ seen **M1** for $\frac{1}{3} \times \pi \times 6^2 \times 12$ seen After 0 scored, **SC1** for consistent use of r = 3 in formula for [hemi]sphere and cone
8(d)(i)	4310 or **FT** 9 × *their* (b)	2	**M1** for $\left(\frac{6}{2}\right)^2$ soi
8(d)(ii)	113 or **FT** $\frac{1}{8}$ × *their* (c)	2	**M1** for $\left(\frac{1}{2}\right)^3$ soi

5. S15/qp21/q4

4	**(a)**	63.6 to 63.62	2	**M1** for πr^2
	(b)	352 to 353	2	**B1** for 161(.2) or 190.9 or 191
	(c)	10	2	**M1** for $\frac{1}{3}\pi 5^2 h$ or $\frac{2}{3}\pi 5^3$

6. W12/qp22/q7

7	(a) (i) 874	3	**M2** for (2) $\pi r^2 + 2\pi r \times 8$ or **M1** for either (2) πr^2 or $2\pi rh$
	(ii) 3070	2ft	**M1** for Figs [(*their* 874 + 150) × 3] or **B1** for $\div 10^4$
	(b) (i) 77 (.0)	1	
	(ii) 500	3ft	**M2** for $\pi R^2 - 4\pi r^2 +$ **4(b)(i)** or **M1** for $\pi R^2 - 4\pi r^2$ or **4(b)(i)**
	(iii) 2410	3	**M2** for $\pi R^2 \times 8 - 4 \times \frac{2}{3} \times \pi \times r^3$ or **M1** for $\pi R^2 \times 8$ or $4 \times \frac{2}{3} \times \pi \times r^3$

7. S11/qp22/q11

11	(a) $\frac{7\pi r^2 H}{9}$	3	B1 for $\frac{2\pi r^2 H}{3}$ and B1 for $\frac{\pi r^2 H}{9}$
	(b) (i) $\sqrt{15^2 + 10^2} = 18(.0)$	2	M1 for $15^2 + 10^2$
	(ii) $62.8 \to 62.9$ or 20π	2	M1 for $2 \times \pi \times 10$
	(iii) $\theta = \frac{62.8 \times 360}{36\pi} = 200°$	2	M1 for $\frac{\theta}{360} \times \pi \times 18 \times 2 =$ their **(ii)**
	(iv) $2760 \to 2770$	3	M1 for $\frac{200}{360} \times \pi \times 18^2 (= 565.5)$ M1 for $30 \times$ their **(ii)** ($= 1884$)

8. W22/qp41/q1 (0580)

1(a)(i)	1580 or 1583 to 1584	**2**	**M1** for $\pi \times 6^2 \times 14$
1(a)(ii)	452 or 452.3 to 452.4...	**2**	**M1** for $\left[\frac{1}{2}\right] \times \frac{4}{3} \times \pi \times 6^3$
1(b)(i)	$7.85 \div 1000$ [= 0.00785]	**M1**	
1(b)(ii)	16[.0] or 15.95 to 15.99	**2**	**FT** {*their* **(a)(i)** + *their* **(a)(ii)**} × 0.00785 evaluated to 3 sig fig or better **M1** for (*their* **(a)(i)** + *their* **(a)(ii)**) × 0.00785
1(c)(i)	16.2 or 16.21 to 16.23	**3**	**M2** for $\frac{2000 - 50 \times \frac{4}{3} \times \pi \times 2^3}{2000}[\times 100]$ or for $\frac{50 \times \frac{4}{3} \times \pi \times 2^3}{2000} \times 100$ or **M1** for $\frac{50 \times \frac{4}{3} \times \pi \times 2^3}{2000}$
1(c)(ii)	6.87 or 6.870 to 6.872	**1**	FT $\sqrt[3]{2000 - \textit{their}\left(50 \times \frac{4}{3} \times \pi \times 2^3\right)}$ evaluated to 3sf or better
1(d)	$\frac{2}{3}$ oe	**4**	**M1** for $[\pi](3R)^2 + [\pi]3R \times 9R$ oe **M1** for $2[\pi]x^2 + 2[\pi]x \times 7x$ oe **M1** for *their* area of cone = *their* area of cylinder seen

9. W21/qp42/q7(a) (0580)

7(a)(i)	$\frac{16\pi}{3}$ or $5\frac{1}{3}\pi$ final answer	2	**M1** for $\frac{1}{2}\times\frac{4}{3}\pi\times2^3$ oe
7(a)(ii)	2.4[0]	4	**B3** for answer in range 2.396… to 2.40… OR **M3** for *their* $\frac{16\pi}{3}+\pi\times2^2\times5.2+\frac{1}{3}\pi\times2^2\times h=\frac{88\pi}{3}$ oe or **M2** for $\frac{88\pi}{3}-$ *their* $\frac{16\pi}{3}-\pi\times2^2\times5.2$ oe or **M1** for $\pi\times2^2\times5.2$ oe or $\frac{1}{3}\pi\times2^2\times h$ oe soi
7(a)(iii)	1 hour 38 min or 1 hour 37.8 min to 1 hour 37.9… min	3	**B2** for 1.63[2…] or 98 [mins] or 97.8 to 97.9…] or **M1** for $\frac{\frac{88\pi}{3}\times620}{35000}$ [× 60] oe

10. S18/qp41/q6 (0580)

6(a)	4.79 or 4.788 to 4.789	3	**M2** for $\sqrt[3]{\frac{230\times3}{2\times\pi}}$ oe or **M1** for $230=\frac{2}{3}\times\pi\times r^3$ oe If 0 scored **SC1** for answer 3.8[0…]
6(b)(i)	8.7[0] or 8.702 to 8.704	3	**M2** for $(300-230)\div(1.6^2\pi)$ or **M1** for $\pi\times1.6^2\times h$
6(b)(ii)	6.4	3	**M2** for $1.6\times\sqrt[3]{\frac{19200}{300}}$ oe or **M1** for sf $\sqrt[3]{\frac{19200}{300}}$ or $\sqrt[3]{\frac{300}{19200}}$ oe or for $\left(\frac{1.6}{r}\right)^3=\frac{300}{19200}$